

Theory Of Knowledge

Prescribed Title 3:

Does it matter if our acquisition of knowledge happens in “bubbles” where some information and voices are excluded? Discuss with reference to two areas of knowledge.

Word Count: 1573

The pursuit of knowledge is often hampered by the bias of excluding or prioritizing significant facts and figures creating a 'bubble'. Acquisition of knowledge, if sought in a 'bubble', impacts the information accumulated, analysis conducted and conclusions drawn in both positive or negative ways, depending on the context. This prescribed title dwells into the discussion of whether obtaining of knowledge occurring in bubbles- where limited information is covered and significant aspects of the truth are removed from consideration- affects the conclusions sought after. This 'bubble' may be beneficial or detrimental to pursuit of knowledge for the two areas of knowledge explored in this essay: history and natural sciences. The pursuit of knowledge should involve the examination and consideration of a wide range of evidence and perspectives in order to arrive at well-informed and nuanced understandings of complex issues. Excluding certain facts or figures and failing to consider multiple viewpoints can lead to incomplete or biased understandings of the subject at hand. The non-consideration of information or a bias formed due to external factors, however, may prove to be beneficial or detrimental to the overall pursuit of knowledge, which is what this essay aims to explore.

Conducting a historical pursuit of knowledge within an isolated 'bubble', resulting from the exclusion of certain perspectives during the declaration of information can have negative consequences. This can be observed in the framing of the Treaty of Versailles, a notable historical example. In the year 1919, the victorious nations of the First World War, namely the USA, UK, and France, formulated the Treaty of Versailles as a consequence for Germany. This treaty included several onerous and unpopular clauses, including the war guilt clause, which held

Germany solely accountable for initiating the war.¹ Widely acknowledged as one of the most unfair and severe treaties, it was formed by the victorious nations by excluding German voices and their struggles during the Paris Peace Conference. This exclusion resulted in the formation of an unfair treaty that further exacerbated the dire circumstances faced by the German population. This example serves to illustrate the formation of the Treaty of Versailles within a ‘bubble’, within which German voices were silenced and therefore excluded, thereby leading to the creation of a treaty with catastrophic repercussions. The harshness of the Treaty of Versailles motivated the Second World War,² thereby causing significant social, economic, and human costs. The animosity harbored by the Germans towards the western nations, coupled with the mutual suspicion arising from Hitler's actions and the rearming movement, resulted in the outbreak of the Second World War.

To consider the other argument, the pursuit of knowledge, if conducted in ‘bubbles’ can prove to be beneficial. There may be occasions in history when the pursuit of knowledge in a more isolated environment has benefited society, for example, The Manhattan Project conducted in the early 20th century. A small group of scientists working on the project were able to develop the first nuclear weapons in a highly isolated and secretive environment.³ The successful development of these weapons played a significant role in ending World War II and helped prevent further loss of life. The pursuit of knowledge, since conducted in a highly isolated environment, formed a

¹ Kirtland, Helen Johns, and Lucian Swift Kirtland. “Jun 28, 1919 CE: Treaty of Versailles.” *National Geographic Society*, 19 May 2022, <https://education.nationalgeographic.org/resource/treaty-versailles-ends-wwi>. Accessed 16 February 2023.

² “The Treaty of Versailles (article).” *Khan Academy*, <https://www.khanacademy.org/humanities/us-history/rise-to-world-power/us-in-wwi/a/the-treaty-of-versailles>. Accessed 16 February 2023.

³ “Security and Secrecy - Nuclear Museum.” *Atomic Heritage Foundation*, <https://ahf.nuclearmuseum.org/ahf/history/security-and-secrecy/>. Accessed 16 February 2023.

‘bubble’ that led to the creation of the most dangerous weaponry significant for many political, social and economical events to come. The conduct of the Manhattan Project in secrecy led to the surrender of Japan in the second world war. Japan played a significant role in World War II, particularly in the Pacific Theater. After Germany surrendered in May 1945, Japan continued to fight against the Allied forces, including the United States, until its own surrender in August 1945. During this time, Japan was led by a militaristic government and had a reputation for committing brutal and indiscriminate acts of violence against civilians, including the use of biological and chemical weapons⁴. Japan was to not surrender from the war as stated by their own engineers, and the US government was well aware of the lengths at which the militaristic Japanese army would go to instead of surrendering. “We would have kept fighting till all the Japanese were killed,” says a Japanese engineer working with the army at the time.⁵ In the final months of the war, the United States dropped atomic bombs on the Japanese cities of Hiroshima and Nagasaki, leading to Japan's surrender and the end of World War II. Although the USA undertook a rather unethical form of attack by bombing civilians, Japan's surrender led to significant reforms making the country a major economic and political power. Which is how, The Manhattan Project, A project conducted in an isolated ‘bubble’, helped end the world war, Japan’s military rule and resulted in world peace.

In the natural sciences, it is essential for researchers to have a variety of views and methods for investigating a certain topic or phenomenon. The formation of a ‘bubble’ for the study of natural sciences may be damaging to the pursuit of knowledge. When scientists work in ‘bubbles’, where

⁴ “Hidden Horrors: Japanese War Crimes in World War II.” *Office of Justice Programs*, <https://www.ojp.gov/ncjrs/virtual-library/abstracts/hidden-horrors-japanese-war-crimes-world-war-ii>. Accessed 16 February 2023.

⁵ Compton, Karl T. “What If the United States Never Dropped the Atomic Bomb?” *The Atlantic*, <https://www.theatlantic.com/magazine/archive/1946/12/if-the-atomic-bomb-had-not-been-used/376238/>. Accessed 16 February 2023.

they only communicate and collaborate with others who share similar ideas and methods, it might result in a lack of critical thinking and stagnation in the understanding of a specific field. Scientists operating in a vacuum may not be exposed to fresh ideas or ways of thinking that could perhaps aid them in making new discoveries or improvements. The research of the safety and effectiveness of hormone replacement treatment (HRT) for menopausal women is a relevant example for this argument. HRT was extensively administered to women for many years to reduce menopause symptoms.⁶ This understanding, however, was based on a small number of observational studies, the majority of which were undertaken by a single group of researchers with a financial stake in the success of HRT, thereby creating a ‘bubble’ in the sample space. These studies were also published in renowned medical publications and widely covered by the media, which helped to the general acceptance of HRT as a safe and effective treatment. Later, however, randomized controlled trials revealed that long-term usage of HRT raised the risk of breast cancer, cardiovascular disease, and other diseases. Due to prior studies conducted in a ‘bubble’ that did not thoroughly investigate long term symptoms or include a diverse set of researchers, this knowledge was not accurate and proven to be unreliable after a more broad research. This example demonstrates how conducting research in a ‘bubble’ can be counterproductive to the pursuit of knowledge, since it might result in a skewed perception of the safety and efficacy of a treatment. In this example, HRT was provided to women for years based on erroneous information concluded in a ‘bubble’, leading to cardiovascular disease, cancer, and osteoporosis.⁷ This also serves as a reminder of the significance of a broad and diverse group of researchers working on a topic, as

⁶ “Hormone replacement therapy (HRT).” *NHS inform*, 3 November 2022, <https://www.nhsinform.scot/tests-and-treatments/medicines-and-medical-aids/types-of-medicine/hormone-replacement-therapy-hrt>. Accessed 16 February 2023.

⁷ Cagnacci, Angelo, and Martina Venier. “The Controversial History of Hormone Replacement Therapy.” *NCBI*, 18 September 2019, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6780820/>. Accessed 16 February 2023.

well as transparency and critical assessment of the studies being undertaken, to guarantee that the results are accurate and objective.

On the other hand, one must consider that conducting research in a ‘bubble’ is not always detrimental to the pursuit of knowledge. In some instances, it may even be advantageous. For instance, in the realm of cancer research researchers may simply concentrate on the genetic mutations that cause certain types of cancer, such as HER2-positive breast cancer, and create medicines that specifically target these alterations. The prioritizing of one type of cancer and the discovery of its treatment creates a research ‘bubble’ that excludes other cancer types and treatments. This provides for a quicker response than researchers looking at cancer as a whole, as they may focus attention on a single aspect of the body and design a cure tailored for that specific form of cancer. This finding, essential to breast cancer research, was made by American oncologist, Dennis J. Slamon, who along with his team at UCLA, chose to use their resources entirely on research for breast cancer. By focusing their study on only breast cancer, they are able to gain a thorough understanding of the disease and further the research for more effective treatments that can increase the survival rates of the patients. In this instance, the researcher's approach of working in a ‘bubble’, by focusing on a single type of cancer and gaining a thorough understanding of it, is proven to be advantageous to the pursuit of knowledge because it enables them to make significant advances in the treatment of that specific disease, which would have been difficult to achieve if they were studying a wider range of conditions. In addition, by having a defined purpose, they can work more efficiently and make the most of the available resources.

To conclude, acquiring knowledge plays a crucial role in fostering independent thinking and debating contentious topics. The idea of ‘bubbles’ provides a useful perspective in comprehending the appropriate boundaries spanning across different disciplines such as history and natural

sciences, as well as between groups of individuals who share similar values, attitudes, and ideas. History can serve as a powerful tool for shaping beliefs and education, but only if all perspectives are taken into account and no one's opinions are distorted. Scientific knowledge, like any other form of knowledge, can be significantly enhanced by continually testing and examining it with new evidence and perspectives. By recognizing the importance of including all voices in the process of acquiring knowledge, individuals can broaden their horizons and gain a more comprehensive understanding of the world.

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